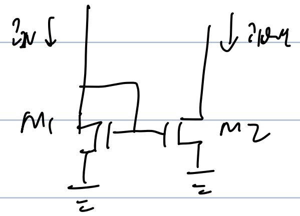
Current Mirror



$$I_{IN} = \frac{k}{2}\frac{W_1}{L_1}(V_{GS1} - V_t)^2$$

$$I_{OUT} = \frac{k}{2}\frac{W_2}{L_2}\cdot(V_{GS2} - V_t)^2$$

$$\Rightarrow \frac{I_{OUT}}{I_{IN}} = \frac{W_2/L_2}{W_1/L_1}$$

$$V_{out,min} = V_{IN} - V_t$$

However, if consider CLM, i.e. $I_{IN} = \frac{k}{2}\cdot\frac{W_1}{L_1}(V_{GS1} - V_t)^2(1+\lambda V_{DS1})$

$$I_{OUT} = \frac{k}{2}\frac{W_2}{L_2}(V_{GS2} - V_t)^2(1+\lambda V_{DS2})$$

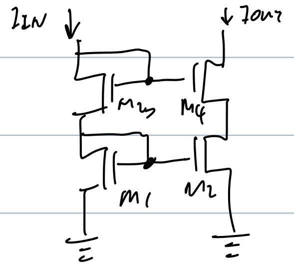
$$\Rightarrow \frac{I_{OUT}}{I_{IN}} = \frac{W_2/L_2}{W_1/L_1}\cdot\frac{1+\lambda V_{DS2}}{1+\lambda V_{DS1}} = \frac{W_2/L_2}{W_1/L_1}\cdot\frac{1+\frac{V_{out}}{V_A}+\frac{V_{IN}}{V_A}-\frac{V_{IN}}{V_A}}{1+\frac{V_{IN}}{V_A}}$$

$$= \frac{W_2/L_2}{W_1/L_1}\cdot\left(1+\frac{\frac{V_{out}}{V_A}-\frac{V_{IN}}{V_A}}{1+\frac{V_{IN}}{V_A}}\right)$$

$$\approx \frac{W_2/L_2}{W_1/L_1}\cdot\left(1+\frac{V_{out}-V_{IN}}{V_A}\right) \quad (V_A \gg V_{IN})$$

如果 $V_{IN} \neq V_{OUT}$，仍会不准。

为了消除这一差异，可以用 cascode current mirror



$$V_{G3} = V_{GS3} + V_{DS1}$$

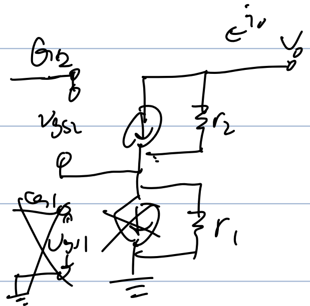
$$\parallel \quad \parallel \Rightarrow \parallel \Rightarrow V_{IN} = V_{OUT}$$

$$V_{G4} = V_{GS4} + V_{DS2}$$

$$V_{out} - (V_{od} + V_t) \geq V_{od}$$

$$\Rightarrow V_{out} \geq 2V_{od} + V_t$$

most accurate current balance.



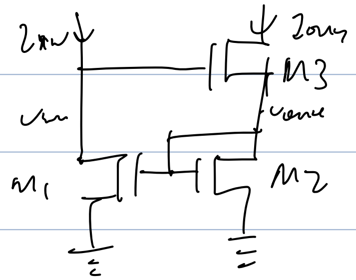
$$v_o = (i_o - g_m V_{gs2})r_2 + i_o r_1$$

$$V_{gs2} = 0 - i_o r_1$$

$$\Rightarrow v_o = (i_o + g_{m2} i_o r_1)r_2 + i_o r_1$$

$$\Rightarrow \frac{v_o}{i_o} = g_{m2} r_1 r_2 + r_1 + r_2 \approx g_{m2} r_{o1} r_{o2}$$

Wilson Current Mirror



$V_{out} - V_{in} = $ const.

Proof:

$V_{in} = V_{GS3} + V_{DS2} = V_{GS2} + V_{DS2}$

$= 2V_{GS2}$

$V_{out} = V_{GS2}$

$\Rightarrow V_{out} - V_{in} = \underline{\underline{V_{GS}}}$

Precise, but not accurate.

High R_{out}.

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